Simulating Cross-Feeding Dynamics Using Proteome-Constrained FBA

1 Protein Allocation Models

Protein Allocation Models extend classical constraint-based models of genome-scale metabolic networks by introducing constraints that reflect condition-dependent patterns of proteome allocation [1,3]. Conveniently, many of the so-called "growth laws" [5] can be straightforwardly integrated into the optimization frameworks where these models are typically used.

The classical formulation of a constraint-based model is:

$$\begin{aligned} \max_{v} \quad & f(v) \\ \text{s.t.} \quad & \\ & Sv = 0 \\ & 0 \leq v \leq U \end{aligned}$$

where v is the flux vector across the network, U are their upper bounds, S is the stoichiometric matrix and f the objective function (eg: biomass production).

To incorporate growth condition-dependent proteome allocation, we must link proteome fractions to the flux variables. This is feasible because most reactions in the network are mediated by proteins (enzymes and transporters). For instance, Alter et al. [1] proposed the following model:

$$\begin{aligned} \max_{v} \quad & f(v) \\ \text{s.t.} \quad & \\ & Sv = 0 \\ & 0 \leq v \leq U \\ & \phi_T = \phi_{T,0} + \omega_T * \mu \\ & \phi_{UE} = \phi_{UE,0} - \omega_{UE} * u \\ & \phi_{AE} = \sum_e \frac{M_e}{k_{cat,e}} * v_e \\ & \phi_T + \phi_{UE} + \phi_{AE} \leq 0.26 \text{ g/gCDW} \end{aligned}$$

These relationships between proteome fractions (ϕ_*) and flux variables are derived from experimental and biochemical data. For instance, Figure 1 illustrates the growth-dependent relationships for ϕ_T (transational related proteome fraction) and ϕ_{UE} ('unused' enzymes proteome fraction).

In short, ϕ_T is assumed to vary linearly with the growth rate ($\mu \in v$), ϕ_{UE} with the uptake of the limiting nutrient ($u \in v$), and ϕ_{AE} (active enzymes proteome fraction) with the linear combination of enzymatic fluxes ($v_e \in v$) where M_e and $k_{cat,e}$ are the effective molecular weight and catalytic

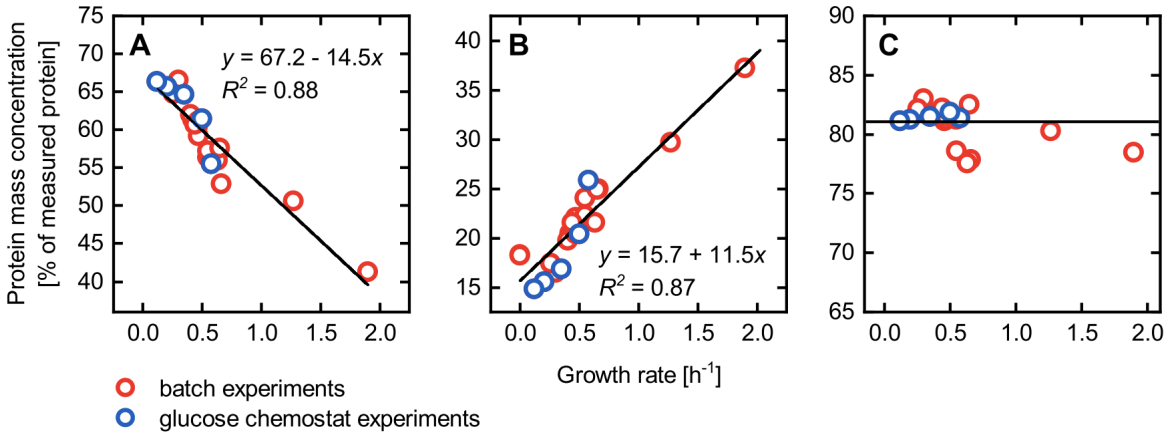


Figure 1: **Protein mass data of distinct protein sectors under diverse growth conditions taken from Schmidt et al. [4]**: (A) All proteins represented by the iML1515 GEM and found in the proteomic data set, comprising the active and unused enzymes sectors. (B) Translational sector covering proteins assigned to the COG (Clusters of Orthologous Groups) class "translation, ribosomal structure, and biogenesis," which are not included in the iML1515. The black lines in panels A and B are linear fits of the data points resulting in the shown equations and coefficients of determination R2 . (C) Sum of the protein mass concentrations shown in panels A and B. The horizontal black line marks the 81% of measured protein mass used to constrain protein availability in the PAM. Glucose chemostat and batch experiments are highlighted in blue and red, respectively. Note that protein concentrations relative to the cell dry weight (cdw) were related to the total protein concentration of 0.32 $g\, g_{cdw}^{-1}$ measured by Schmidt et al [4]. **Both figure and caption taken from [1].**

constant of enzyme e, respectively. The total proteome fraction is bounded by an experimentally determined value. See further details at [1].

It is convenient that these constraints can be expressed as linear relations, allowing the use of efficient linear programming algorithms.

2 Coexistence model

We can extend the proteome allocation model described above to include two distinct stoichiometric matrices, one for each auxotrophic strain. Figure 2 descirbe the model general layout. A key feature is that the two networks interact only through the medium; no direct interaction between strains is modeled. This allows independent control of the exchange reactions for each strain. This can be achieved by adding extra reactions and metabolites that model the mass flux to/from the 'medium', and to/from the 'universe'.

3 DFBA: Coexistence in a Chemostat

Proteome allocation models show significantly improved ability to predict the physiological behavior of E. coli cultures compared to standard stoichiometric models [1]. This motivates evaluating their performance in the context of coexistence. To that end, we use dynamic flux balance analysis (DFBA) to simulate a chemostat culture of E. coli auxotrophs engineered to engage in forced cross-feeding of essential amino acids.

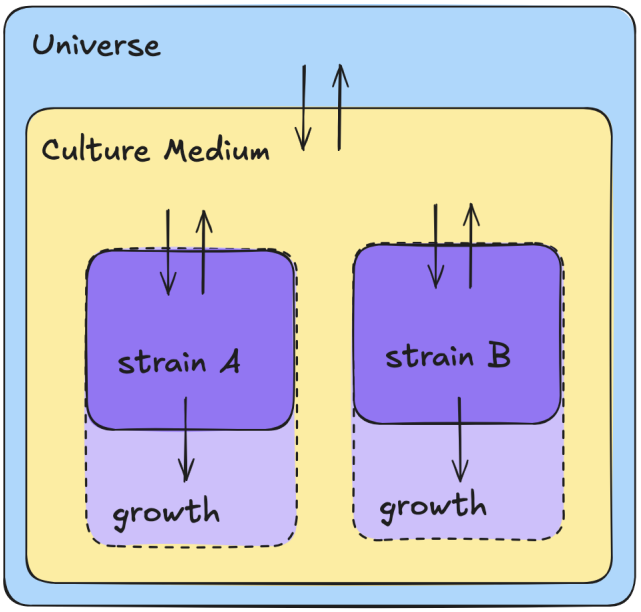


Figure 2: **Schema of a mixed model**: Arrows indicate mass fluxes. Note that strains can only interact by its influence with the medium.

For simplicity, the simulations only track the dynamics of the cell densities (X_*) and the single carbon source (s_{glc}). The carbon source uptake is modeled using the standard Michaelis-Menten kinetics:

$$U_{\text{intake glc}} = V_{glc}^{max} * \frac{s_{glc}}{KM_{glc} + s_{glc}}$$

note that U_{glc} will be common for all strains.

Any component of the medium supplied by the 'universe' is considered infinitely available, and its dynamics are therefore neglected. Additionally, nutrients produced by the strains are assumed to be non-accumulating. That is, at any moment, each strain can consume only what the other is producing, in proportion to their respective cell densities. For a metabolite k we have:

$$v_{\text{intake k}}^{\text{strain A}} \leq v_{\text{production k}}^{\text{strain B}} * \frac{X^{\text{strain B}}}{X^{\text{strain A}}}$$

With this setup, we can simulate the dynamics of a chemostat culture:

$$\begin{aligned}
\frac{X^*(t+1) - X^*(t)}{\Delta t} &= (\mu - D) * X^*(t) \\
\frac{s_{glc}(t+1) - s_{glc}(t)}{\Delta t} &= (c_{glc} - s_{glc}(t)) * D + \sum_i (v_{\text{production k}}^i - v_{\text{intake k}}^i) * X^{\text{strain i}}
\end{aligned}$$

where the sum is over all strains and X^* is the cell density of a given strain. The fluxes μ and v_*^* are computed by solving an FBA problem contextualized by the current concentration and production of nutrients.

4 Results

We simulate co-cultures of two E. coli auxotrophs: ΔIle, which cannot produce isoleucine, and ΔMet, which cannot produce methionine.

Because the appropriate objective function is unclear, we evaluate several alternatives. Each objective function is a linear combination of fluxes and can be parameterized by the choice of coefficients.

We used the follow linear objective function:

- **ΔIle.egoist.max.yield**: We use coefficients spanning different orders of magnitude to setup a hierarchical optimization. In this case the objective function will be:

$$\begin{aligned} f(v) =& 10^4 \mu^{\Delta Ile} + 10^2 \mu^{\Delta Met} \\ & - 10^{-4} v^{\Delta Ile}_{\text{intake gluc}} - 10^{-4} v^{\Delta Met}_{\text{intake gluc}} \\ & - 10^{-4} v^{\Delta Ile}_{\text{intake Ile}} - 10^{-4} v^{\Delta Met}_{\text{intake Met}} \end{aligned}$$

  This formulation tends to maximize both biomass production rates while minimizing nutrient intake. However, due to the large differences in coefficient magnitudes, the optimizer will prioritize the higher-weighted terms.

- **ΔMet.egoist.max.yield** Analogous to the previous one but changing the biomass coeffitients so mutant ΔMet is prioritized. The rest is identical to the first objective function.

$$f(v) = 10^2 \mu^{\Delta Ile} + 10^4 \mu^{\Delta Met} \dots$$

- **ΔIle.egoist.only**: This time the only biomass flux being optimized (maximized) is the biomass rate of the ΔIle mutant. The rest is identical to the first objective function.

$$f(v) = 10^4 \mu^{\Delta Ile} + 0.0 \mu^{\Delta Met} \dots$$

- **ΔMet.egoist.only**: Analogous to the previous one but for the ΔMet mutant. The rest is identical to the first objective function.

$$f(v) = 0.0 \mu^{\Delta Ile} + 10^4 \mu^{\Delta Met} \dots$$

- **non-competitive.max.yield** In this case, both biomasses have equal weight in the optimization. The rest is identical to the first objective function.

$$f(v) = 10^4 \mu^{\Delta Ile} + 10^4 \mu^{\Delta Met} \dots$$

Figure 3 shows representative dynamics for cell densities and residual glucose concentration. In most cases, the dynamics are smooth, but at high dilution rates they may become unstable (see bottom left panel). These unstable simulations are excluded from analysis.

Another observation is that all simulations with **Δ*.egoist.only** objective functions failed to achieve coexistence. This is likely because maximizing the growth of only one strain (the egoist) results in no flux being allocated to the other strain's biomass, leading to its eventual washout. This, in turn, limits the egoist's access to the essential metabolite it depends on, preventing it from maintaining growth above the dilution rate. The top-right panel of Figure 3 illustrates this scenario.

Figures 4 and 5 show the results for the simulations that reached a stady state. From the figures, we can make the following observations:

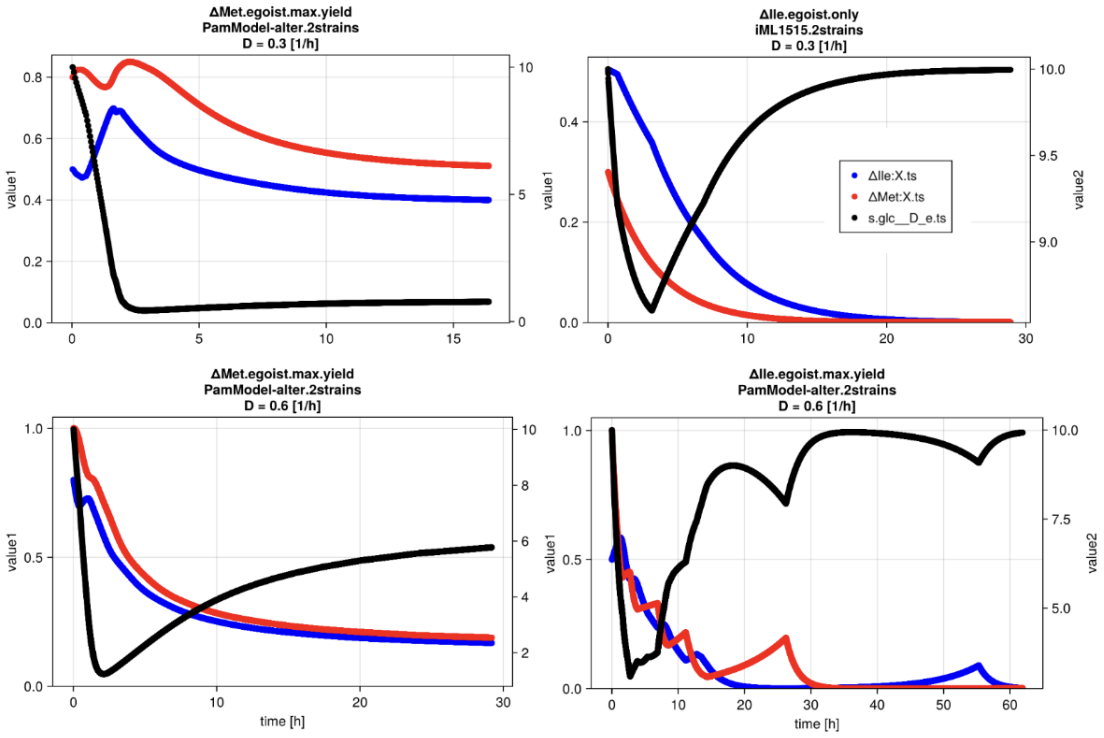


Figure 3: **Representative dynamic simulations**: The plots show the evolution of cell density (right y-axis) and glucose concentration in the medium (left y-axis) under different conditions.

- The total culture biomass decreases as the dilution rate (D) increases ((Figure 4)).
- At higher dilution rates, the steady-state cell densities of both mutants tend to converge ((Figure 4)).
- The selection of objective function affects the initial difference of steady state cell density at low dilution rate.
- Biomass yield (g biomass per mmol glucose) declines with increasing D (Figure 5)

5 Further Considerations

- Identify interesting questions that can be addressed using genome-scale metabolic models, which cannot be captured by simpler dynamic models.
- Consider incorporating the effect of antibiotics, as described by another "growth law" [2], and investigate its impact on coexistence. For example, what happens if only one strain is sensitive while the other is resistant?
- From an experimental perspective, all these experiments are feasible using the DIY chemostat setup.

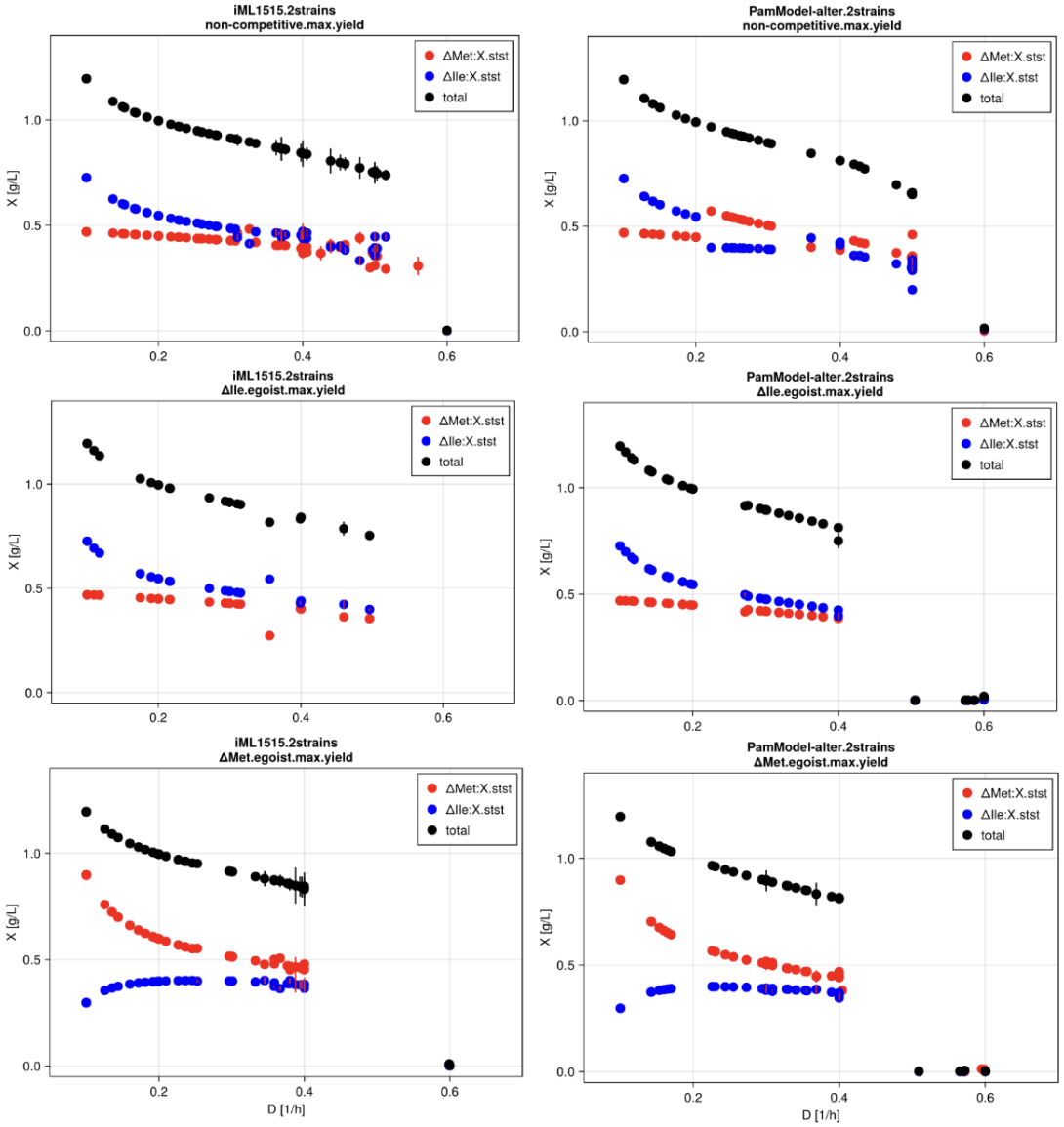


Figure 4: **Steady-state cell densities**: Results are shown for both the proteome allocation model (PamModel-alter.2strains) and the baseline model (iML1515.2strains) without proteome allocation constraints. Only simulations where cell densities remained stable (± 20%) during the final 10% of the time window are included. Error bars indicate this variation.

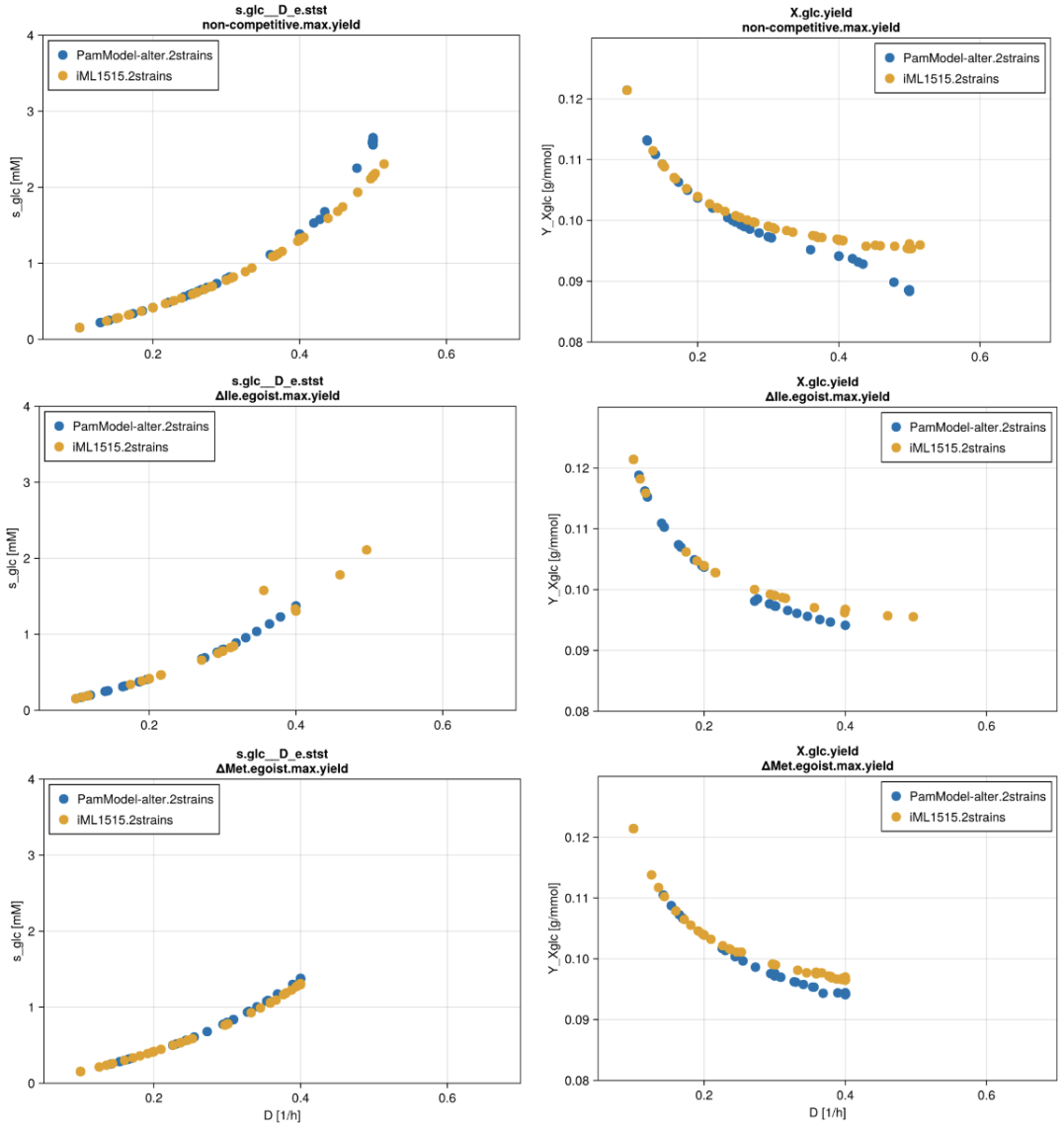


Figure 5: **Steady-state glucose levels and biomass yield**: This figure parallels Figure 4 but reports residual glucose concentration (mM) and biomass yield (g/mmol glucose).

References

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